IT-SIMPLERA Institute

Cyber Security Analyst Internship Program

WEEK 02 ASSIGNMENT

Computer Networking Fundamentals & Network Configuration

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Introduction

Computer networking is the foundation of modern communication systems, enabling computers and devices to exchange data efficiently over local and global networks. Understanding networking concepts is essential for cybersecurity professionals because secure communication depends on properly configured and managed networks.

This assignment focuses on identifying network configuration details, using basic networking commands, comparing networking devices and Internet Protocol versions, and understanding the OSI Model. These activities help develop practical troubleshooting skills and strengthen the theoretical knowledge required in cybersecurity.

Task 1: Network Information

1. IPv4 Address: 192.168.199.91

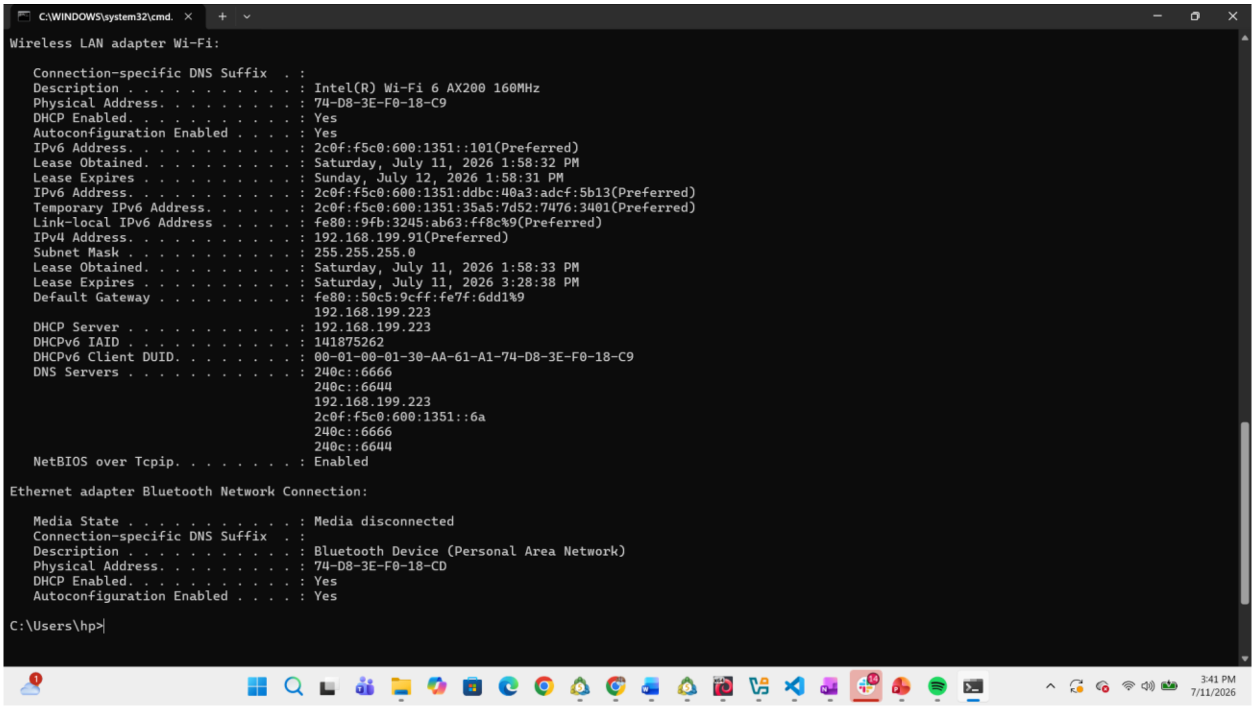
An IPv4 address is a unique numerical identifier assigned to a device on a network. It enables the device to communicate with other devices and access network resources using the Internet Protocol (IPv4).

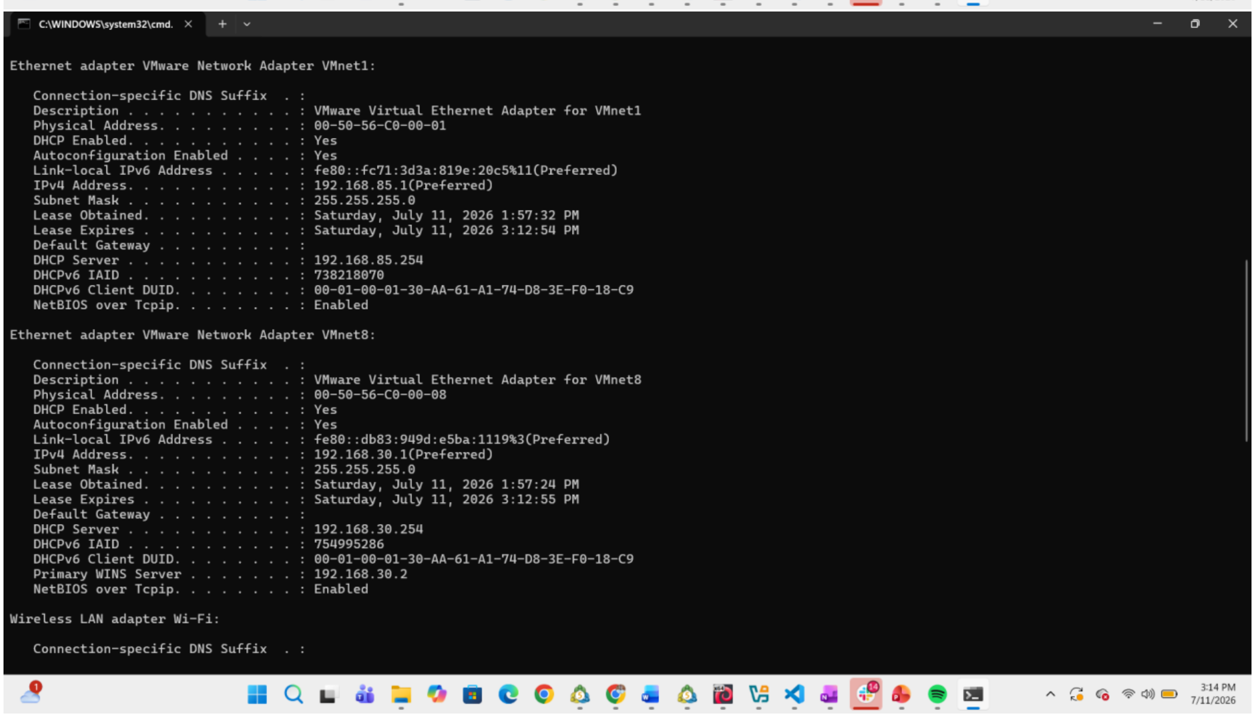
2. MAC Address: 74-D8-3E-F0-18-C9

A MAC (Media Access Control) address is a unique hardware address assigned to a network interface card (NIC). It is used to identify a device on a local area network (LAN).

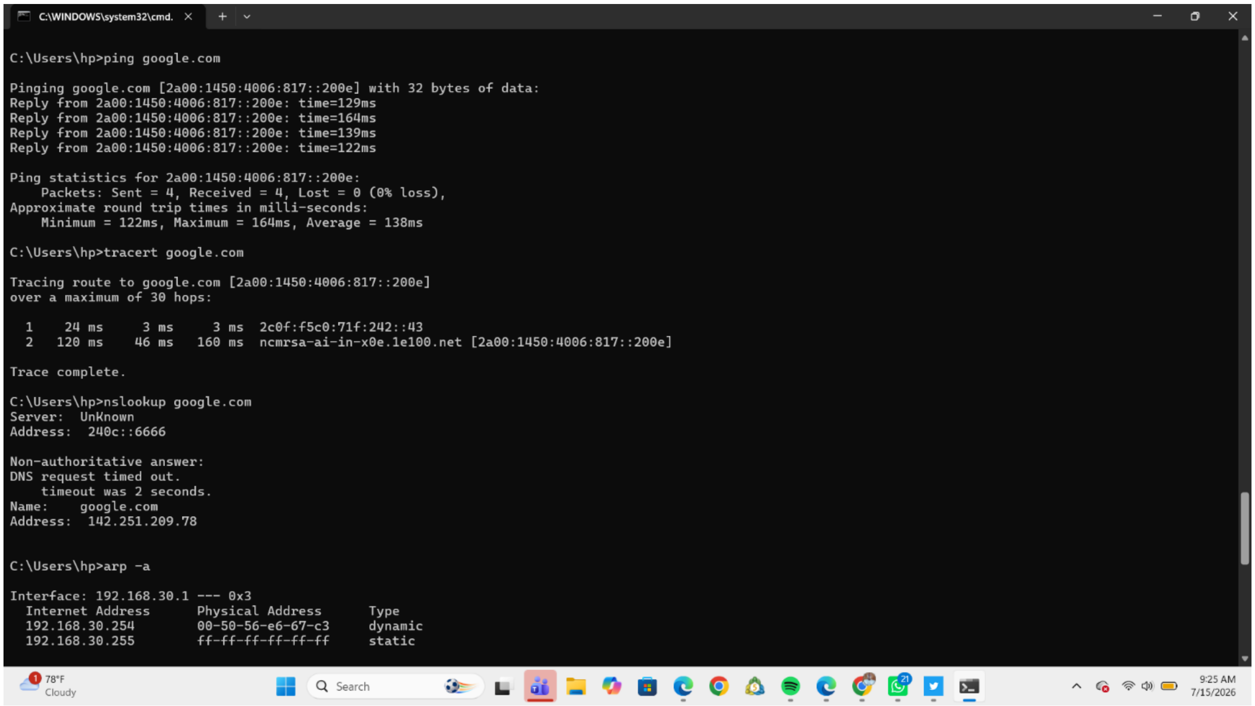
3. Default Gateway: 192.168.199.223

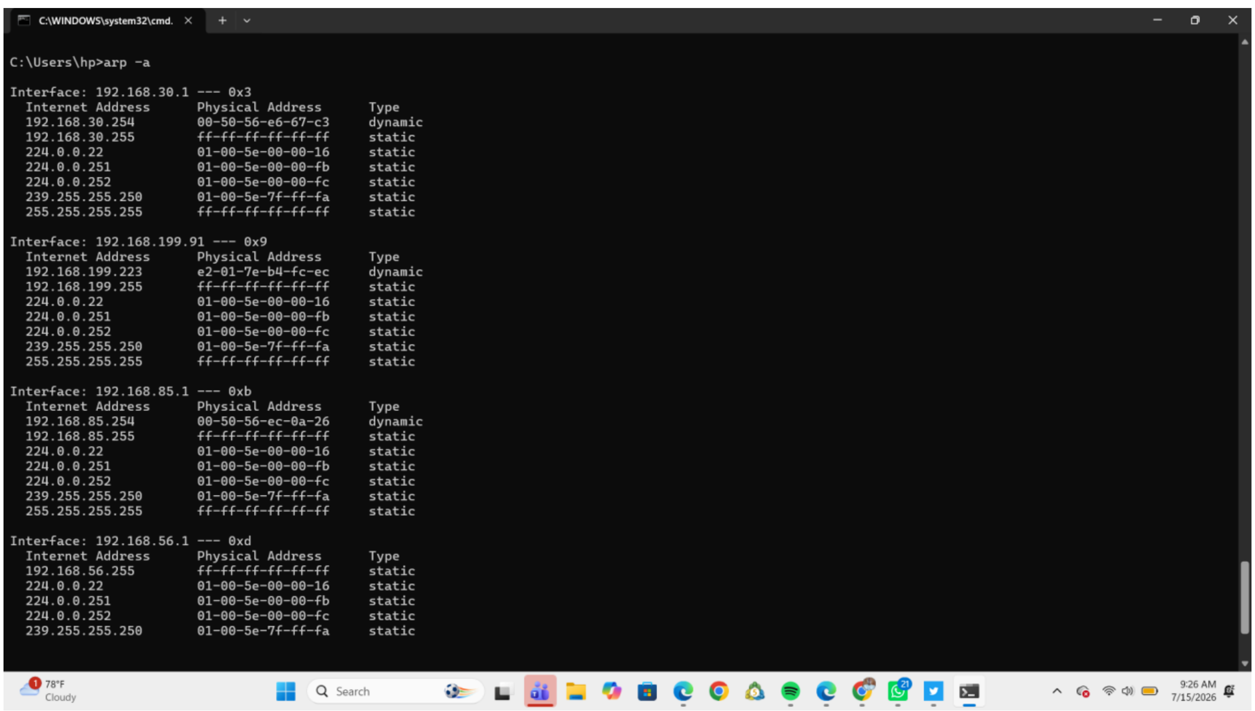
The Default Gateway is the network device (usually a router) that forwards data from the local network to other networks, including the Internet. It serves as the exit point for network traffic.





Task 2: Basic Networking Commands





Command	Purpose
`ipconfig /all`	Displays detailed network configuration information, including the computer's IP address, MAC address, DNS servers, and default gateway. It is used to troubleshoot network connectivity issues.

Command	Purpose
`ping google.com`	Tests connectivity between the computer and a remote host by sending ICMP Echo Request packets. It also measures the response time and checks for packet loss.
`tracert google.com`	Displays the path (route) that packets take from your computer to the destination host. It helps identify where delays or network problems occur.
`nslookup google.com`	Queries the Domain Name System (DNS) to translate a domain name into its corresponding IP address. It is commonly used to verify DNS resolution.
`arp -a`	Displays the Address Resolution Protocol (ARP) cache, showing the mapping between IP addresses and MAC addresses on the local network. It helps troubleshoot local network communication.

Task 3: Router vs. Switch

Feature	Router	Switch
Primary Function	Connects different networks and forwards data between them.	Connects devices within the same local area network (LAN).
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer) *(some advanced switches can operate at Layer 3)*
Address Used	Uses **IP addresses** to route packets.	Uses **MAC addresses** to forward frames.
Coverage	Connects multiple networks, such as a home network to the Internet.	Connects multiple devices within a single network.
Broadcast Domain	Separates broadcast domains, reducing unnecessary network traffic.	Devices connected to the same switch are typically in the same broadcast domain unless VLANs are configured.
Common Use Cases	Home routers, office gateways, Internet Service Provider (ISP) networks.	Connecting computers, printers, servers, and access points in offices or schools.

Conclusion

A **router** is used to connect different networks and direct data between them using IP addresses, while a **switch** connects devices within the same network using MAC addresses. Both are essential networking devices but serve different purposes.

Task 4: IPv4 vs. IPv6

Feature	IPv4	IPv6
Address Length	32-bit address	128-bit address
Address Format	Four decimal numbers separated by dots	Eight groups of hexadecimal numbers separated by colons
Example Address	**192.168.199.91**	**2001:0db8:85a3:0000:0000:8a2e:0370:7334**
Number of Addresses	Approximately 4.3 billion addresses	Approximately 340 undecillion (3.4×10^{38}) addresses
Configuration	Can be configured manually or using DHCP	Supports automatic configuration (SLAAC) and DHCPv6
Security	IPsec is optional	IPsec support is built into the protocol (though its use is optional)
Broadcast Support	Supports broadcast, multicast, and unicast	Does **not** support broadcast; uses multicast and anycast instead
Header Size	Variable header (20–60 bytes)	Fixed header (40 bytes), making packet processing more efficient

Conclusion

IPv4 is the older Internet Protocol with a limited address space, while IPv6 is the newer version designed to provide a vastly larger number of addresses, improved efficiency, and better support for modern networking requirements.

Task 5: OSI Model

Layer (Top to Bottom)	Primary Function	Common Protocol	Networking Device
7. Application	Provides network services directly to end-user applications.	HTTP	Proxy Server
6. Presentation	Translates, encrypts, and compresses data between the application and network.	SSL/TLS	Gateway
5. Session	Establishes, manages, and terminates communication sessions between devices.	NetBIOS	Gateway
4. Transport	Ensures reliable data delivery, error checking, and flow control.	TCP	Firewall

Layer (Top to Bottom)	Primary Function	Common Protocol	Networking Device
3. Network	Determines the best path for data and routes packets between networks.	IP (IPv4/IPv6)	Router
2. Data Link	Transfers data between devices on the same network and uses MAC addresses.	Ethernet (IEEE 802.3)	Switch
1. Physical	Transmits raw bits over the physical transmission medium.	Ethernet (Physical Layer Standard)	Hub/Cables/Repeater

OSI Model Summary

- **Layer 7 – Application:** Provides services for user applications such as web browsing and email.
- **Layer 6 – Presentation:** Formats, encrypts, and compresses data.
- **Layer 5 – Session:** Creates, maintains, and closes communication sessions.
- **Layer 4 – Transport:** Ensures reliable and complete data transfer using protocols such as TCP.
- **Layer 3 – Network:** Routes packets between different networks using IP addresses.
- **Layer 2 – Data Link:** Delivers frames within the same network using MAC addresses.
- **Layer 1 – Physical:** Transmits binary data through physical media such as cables or wireless signals.

Challenges Faced

During this assignment, I encountered a few challenges while identifying the correct network adapter because my computer contained multiple VMware virtual network adapters in addition to the Wi-Fi adapter. I also learned how to interpret the output of networking commands such as **ipconfig**, **ping**, **tracert**, **nslookup**, and **arp**. Completing these tasks improved my understanding of network configuration and troubleshooting techniques.

Conclusion

This assignment provided valuable practical experience in computer networking fundamentals. I successfully identified important network configuration details, executed essential networking commands, compared routers and switches, distinguished between IPv4 and IPv6, and studied the seven layers of the OSI Model. The knowledge gained from this assignment has strengthened my networking skills and established a solid foundation for future cybersecurity and network administration tasks.